

State Spread-Pricing Bans and Medicaid Generic Drug Spending: What Public Data Can and Cannot Identify

Structured Abstract

Objective. To evaluate whether 2018–2024 state prohibitions on pharmacy benefit manager (PBM) spread pricing reduced Medicaid managed care organization (MCO) generic drug spending using publicly available data, to explicitly bound the cross-state-average spread fraction consistent with that evidence, and to test whether bans are empirically embedded in a broader pharmacy cost-containment bundle that public data cannot disentangle.

Data Sources and Study Setting. CMS State Drug Utilization Data (SDUD), National Average Drug Acquisition Cost (NADAC), and Medicaid enrollment files, 51 U.S. jurisdictions, 2016Q1–2024Q4. A new state-year panel codes five bundled cost-containment actions (MAC list updates, preferred drug list refreshes, NADAC alignment, pharmacy carve-outs, single-PBM transitions).

Study Design. Staggered difference-in-differences using Callaway–Sant’Anna (primary) and two-way fixed effects (benchmark), with a Goodman–Bacon decomposition to isolate clean 2×2 comparisons. A triple-difference (generic versus brand within MCO) and a pre-registered negative control (MCO versus fee-for-service within state) test the mechanism. A bundled-action state-year panel tests whether treated states disproportionately pursued other levers. Structural bounds use the 2018 Ohio audit spread as a calibration prior.

Data Collection/Extraction Methods. Spending aggregated to state-quarter from SDUD and merged with NADAC by National Drug Code. Treatment and bundled-action dates drawn from NASHP, KFF, HMA, and state Medicaid bulletins; every coded cell retains a source URL.

Principal Findings. The Callaway–Sant’Anna simple aggregate is -0.054 log points (SE 0.477); TWFE -0.058 (SE 0.231). 12.6 percent of the TWFE estimate is attributable to contaminated late-versus-early comparisons; the Bacon-clean weighted average is -0.118 . The triple-difference is -0.117 (SE 0.120) with a clean brand placebo. The pre-registered negative control fails: fee-for-service generic spending declines by -0.252 (SE 0.205), larger than the MCO estimate, in every robustness specification. Treated states take approximately half an additional bundled action per state-year ($+0.507$, SE 0.107, $p < 0.001$). The triple-difference implies a cross-state-average generic spread fraction of 11.0 percent (95% CI -12.5 to $+29.6$ percent) against a 31.4 percent Ohio-audit prior.

Conclusions. Public SDUD/NADAC data cannot distinguish the spread-pricing mechanism from contemporaneous bundled state pharmacy cost containment. The MCO point estimate is consistent with anywhere from substantial savings (lower edge of the triple-difference CI approaches -30 percent) to zero. The decisive test awaits claim-level Medicaid T-MSIS data or the spread-reporting infrastructure coming online in rating periods beginning November 2025.

Keywords: pharmacy benefit managers; spread pricing; Medicaid managed care; negative control; cost containment; difference-in-differences

1. Introduction

Between 2018 and 2024, at least 24 U.S. states prohibited pharmacy benefit manager (PBM) spread pricing in Medicaid managed care.^{7,8} The wave began in Ohio, where an August 2018 audit of 39

million Medicaid drug transactions documented a 31.4 percent spread on generic drugs.¹ Similar audits in Kentucky, Michigan, and Massachusetts followed, and the U.S. Federal Trade Commission’s 2024 and 2025 interim reports documented large spreads and affiliated-pharmacy markups at the three largest PBMs.^{2,3} The Consolidated Appropriations Act of 2026 (CAA 2026), signed February 3, 2026, codifies the expectation at the federal level by prohibiting PBM compensation from spread pricing in Medicare Part D effective January 2028.⁴

This paper provides the first peer-reviewed quasi-experimental evaluation of the 2018–2024 state ban wave using publicly available CMS data. Using a balanced 51-state quarterly panel over 2016Q1–2024Q4, we estimate the effect of state bans on log MCO generic drug spending per MCO enrollee and on a battery of secondary outcomes designed to probe the mechanism. Identification relies on staggered difference-in-differences with Callaway–Sant’Anna as the primary estimator²⁰ and a Goodman-Bacon decomposition of the TWFE benchmark²¹ to isolate clean 2×2 comparisons. A pre-registered within-MCO triple-difference (generic versus brand) tests whether any detected effect is generic-concentrated as Ohio audits predict, and a pre-registered MCO-versus-fee-for-service (FFS) negative control tests whether any detected state-level decline is PBM-specific. Because FFS pharmacy reimburses dispensing pharmacies directly with no PBM intermediary retaining spread, an FFS effect should be zero under the clean-mechanism hypothesis.

We commit this paper to an honest reframe. The data cannot cleanly identify the spread-pricing mechanism from public CMS files. Three findings establish the reframe. First, the main DiD is a precise null (Callaway–Sant’Anna -0.054 log points, SE 0.477; TWFE -0.058 , SE 0.231), and the triple-difference tightens to -0.117 (SE 0.120) with a clean brand placebo — directionally consistent with the Ohio prior but not significant at conventional levels. Second, the pre-registered MCO-versus-FFS negative control fails: FFS generic spending in treated states declines by -0.252 log points (SE 0.205), larger than the MCO estimate, surviving every robustness specification in a pre-registered measurement drill. Third, we construct a new state-year panel of five bundled pharmacy cost-containment actions and show that states that enacted spread-pricing bans take approximately half an additional bundled action per state-year beyond what state and year fixed effects predict ($+0.507$, SE 0.107, $p < 0.001$), concentrated in MAC-list updates, NADAC alignment, and PDL refreshes. The pattern is diagnostic: spread-pricing bans co-occur with contemporaneous bundled state pharmacy cost containment that affects both MCO and FFS channels through shared administrative infrastructure.

Under this reframe, the MCO point estimate is consistent with anywhere from substantial savings to zero. A structural decomposition that treats the 31.4 percent Ohio generic spread as a calibration prior places the implied cross-state-average spread fraction from the triple-difference at 11.0 percent (95 percent CI -12.5 to $+29.6$ percent) — a band that statistically rejects the Ohio prior but does not rule out a cross-state-average spread of 20 to 30 percent. The decisive test awaits claim-level Medicaid T-MSIS data or the spread-reporting infrastructure coming online under the November 2023 Medicaid Managed Care final rule¹⁰ in rating periods beginning November 2025.

Our contribution is four-fold. First, we deliver the first cross-state quasi-experimental evaluation of the 2018–2024 spread-pricing ban wave using public data. Second, we document a substantive pre-registered negative-control failure and show, through a measurement drill, that the failure is real rather than mechanical. Third, we build a state-year panel of bundled pharmacy cost-containment actions — a new empirical object that future researchers and state Medicaid analysts can reuse — and we use it to test the cost-containment bundle hypothesis directly. Fourth, we deliver a data-infrastructure finding with direct relevance to federal evaluators implementing the CAA 2026 Medicare Part D ban: public SDUD/NADAC aggregates cannot distinguish mechanism-specific

PBM policy effects from bundled state pharmacy cost containment, and the infrastructure needed to do so must be built before the 2028 effective date arrives.

The remainder of the paper is organized as follows. Section 2 provides institutional background. Section 3 reviews related literature. Section 4 describes the data. Section 5 lays out the empirical strategy. Section 6 presents results. Section 7 discusses implications and limitations. Section 8 concludes.

2. Background

Medicaid pharmacy benefit structure. Medicaid is the largest U.S. pharmacy benefit payer, with gross drug spending of approximately \$80 billion in fiscal year 2023.⁵ Benefits are delivered through two channels. In FFS pharmacy, the state Medicaid agency reimburses dispensing pharmacies directly at acquisition cost plus a dispensing fee, typically benchmarked to NADAC and supplemented by state maximum allowable cost (MAC) lists for generics. FFS pharmacy is structurally a pass-through — there is no intermediary retaining a spread. In MCO pharmacy, the state contracts with managed care plans that subcontract with PBMs to administer the drug benefit; approximately 64 percent of Medicaid drug spending flowed through the MCO channel in FY2023.⁵

PBM spread pricing and its generic concentration. Under spread pricing, the PBM charges the MCO one amount for a dispensed drug and pays the pharmacy a lower amount; the PBM retains the wedge. The wedge is heavily concentrated in generics: the Ohio audit found a 31.4 percent spread on generics versus 0.8 percent on brands.¹ Mattingly et al.⁶ quantified this directly in Medicare Part D for 45 high-utilization generics, finding that PBMs retained 40.8 percent of mean per-claim spending.

The 2018–2024 state ban wave. Ohio acted first (administrative directive effective 2019Q1 requiring transparent pass-through compensation). New York followed in 2019Q4, Arkansas in 2020Q3, and Kentucky, Pennsylvania, Virginia, and Maryland in 2021; North Carolina, Massachusetts, Florida, Idaho, and Vermont adopted in 2021–2024. Ten additional states — Georgia, Iowa, Kansas, Louisiana, Michigan, Minnesota, Mississippi, North Dakota, New Jersey, and Texas — had contract-based prohibitions in force by 2018Q1, embedded in standard MCO contract boilerplate before Ohio’s high-profile audit.^{7,8} These ten “early adopter” states form a 2018Q1 mass cohort in our panel; their treatment timing is fuzzier because bans were embedded in contract renegotiation cycles rather than single-date actions. We evaluate them together as the mass cohort and in a policy-window sensitivity that drops them. Twenty-one of the 24 states have in-window effective dates (2018–2024); California, Colorado, and Idaho’s 2025/2026 effective dates are recoded as never-treated. An appendix table reconciles the 24-state count with the “21 in-window” and “20 treated in the main spec” (after Vermont’s all-FFS exclusion) numbers across the text.

Related policies within the bundle. Several states pursued more radical structural alternatives during the study window: California’s January 2022 Medi-Cal Rx carve-out moved MCO pharmacy volume into a single FFS-equivalent administrator (Magellan); Ohio’s October 2022 Single PBM (SPBM) model routed all MCO pharmacy claims through a state-contracted PBM (Gainwell). West Virginia carved out in 2017, Kentucky moved to a single PBM in 2021, New York launched NYRx in April 2023, and Louisiana contracted Magellan in October 2023. States also updated MAC lists, refreshed PDLs, and aligned MCO reimbursement to NADAC benchmarks during the same window — the “bundle” we test directly in Section 6.4.

Federal regulatory context. CMS issued guidance in May 2019 encouraging states to prohibit

spread pricing.⁹ The November 2023 Medicaid Managed Care final rule requires PBM subcontractors to separately report the spread amount beginning in rating periods on or after November 19, 2025¹⁰ — a data-infrastructure step that will for the first time produce state-level aggregate measures of the MCO-pharmacy wedge. The FTC’s Section 6(b) investigation produced two interim reports documenting anticompetitive practices.^{2,3} CAA 2026 bans PBM spread-pricing compensation in Medicare Part D effective January 2028 and directs GAO PBM compensation studies.⁴ CAA 2026 does not affect the 2016–2024 empirical window; it is relevant here only as the regulatory backdrop that makes the data-infrastructure finding policy-salient.

Why a pre-registered negative control. Separating the ban-specific effect from other contemporaneous state pharmacy cost-containment activity is the central identification challenge. Because FFS pharmacy is structurally immune to PBM spread pricing, a pre-registered FFS-channel negative control should deliver a zero estimate under the clean-mechanism hypothesis. A non-zero FFS effect indicates either a parallel-trends violation, a different mechanism, or confounding with bundled state pharmacy policy operating outside the MCO-PBM channel. This logic motivates both the pre-registration and the reframe.

3. Related Literature

Spread pricing audits and mechanism evidence. The state audit literature provides strong priors for the existence of PBM spread pricing as a margin-retention practice. Ohio’s August 2018 report — a 31.4 percent spread on generics and \$208 million retained — is the canonical reference.¹ Kentucky, Michigan, New York, and Massachusetts found similar patterns.^{11,12} Werble¹³ provided an early descriptive analysis; Mattingly, Hyman, and Bai¹⁴ reviewed the institutional structure. The audit literature documents existence at specific states and times but does not document that *banning* spread pricing produces measurable aggregate state-quarter spending reductions detectable by a policy evaluator. This paper tests that implication.

Medicaid pharmacy policy evaluation. Dranove, Ody, and Starc¹⁵ used a dose-of-managed-care design to estimate MCO pharmacy penetration effects on Medicaid drug spending. SDUD is well-validated for state-level Medicaid pharmacy analysis.¹⁶ Bendicksen and Kesselheim¹⁷ frame the MCO-PBM spread as a principal-agent problem. No peer-reviewed paper has used SDUD to evaluate the 2018–2024 state spread-pricing ban wave. A Congressional Budget Office analysis projects that a federal Medicaid spread-pricing ban would reduce federal outlays by \$1.1 billion over 2024–2033 using audit-derived spread magnitudes rather than post-ban empirical data. A Milliman actuarial evaluation for the Ohio Department of Medicaid uses internal Ohio data that separately attributes dispensing fees, PBM administrative fees, and retained spread — data SDUD does not expose.

Negative controls and placebo identification. Lipsitch, Tchetgen Tchetgen, and Cohen¹⁸ formalized negative controls in epidemiological inference. Roth, Sant’Anna, Bilinski, and Poe¹⁹ summarize negative-control best practices in staggered DiD with explicit treatment of within-panel contrasts (generic versus brand; MCO versus FFS; eligible versus ineligible) as sharper identification tools than relying on parallel-trends alone. The natural negative control for a spread-pricing ban is the MCO-versus-FFS contrast within state. Its pre-registration and subsequent failure are the empirical fulcrum of this paper.

Staggered DiD methodology. We use Callaway and Sant’Anna²⁰ as the primary estimator and TWFE as a benchmark. Goodman-Bacon²¹ and Sun and Abraham²² formally documented the heterogeneous-treatment-effects bias that contaminates TWFE in staggered adoption. With 20

treated states entering at different quarters in 2018–2024, the setting is canonical for CS-DiD. For pre-trend sensitivity we use Rambachan and Roth23 relative-magnitudes honest-DiD bounds.

4. Data

We combine four public data sources to construct a balanced state-quarter panel (36 quarters $\times$ 51 state-equivalents including DC = 1,836 observations) and a new state-year panel of bundled cost-containment actions ($51 \times 9 = 459$ cells).

State Drug Utilization Data (SDUD). SDUD reports the number of prescriptions, units dispensed, and total amount reimbursed for each National Drug Code (NDC) in each state-quarter, separately for FFS and MCO. We use annual files for 2016 through 2024 from data.medicaid.gov. Records with fewer than 11 prescriptions are suppressed by CMS. We exclude U.S. territories. SDUD is the standard public source for state-level Medicaid pharmacy research and has been used in prior peer-reviewed Medicaid pharmacy studies, including Mahendraratnam, Dusetzina, and Farley’s analysis of state-level Medicaid prescription utilization and reimbursement.²⁴

National Average Drug Acquisition Cost (NADAC). NADAC, calculated monthly for CMS, approximates the average invoice price pharmacies pay for drugs. We use weekly reference files for 2016–2024 and compute quarterly mean NADAC per unit for each NDC. Approximately 85–90 percent of SDUD NDCs by dispensing volume match to a NADAC record; the residual 10–15 percent is predominantly physician-administered brand drugs, compounded products, and newly launched drugs. Unmatched records are excluded from NADAC-based markup calculations but retained in total spending aggregates. A match-rate diagnostic by treatment status is in Appendix Table A1.

Medicaid enrollment. MCO and FFS enrollment counts are drawn from CMS MBES and the CMS Share of Medicaid Enrollees in Managed Care annual reports. MCO = comprehensive managed care (excluding limited-benefit and PCCM-only); FFS = total minus MCO. Five Michigan state-quarters produce small negative residuals (magnitudes under 600 enrollees) from CMS rounding and are winsorized to zero; dropping Michigan entirely does not change any main estimate (Appendix Table A2, row “drop MI”).

Treatment panel. Ban dates are drawn from KFF State Health Facts, the NASHP State PBM Legislation Tracker, the HMA 2024 50-State Medicaid Pharmacy Survey, and state bulletins.^{7,8} Twenty-four states code as enacted; 21 have in-window effective dates; Vermont is flagged (all-FFS pharmacy channel) and recoded as never-treated in the main spec; Tennessee is excluded (single-PBM structure); California, Colorado, and Idaho (2025–2026 effective dates) are recoded as never-treated. Appendix Table A3 lists every state with vehicle (statute / contract / directive), adoption quarter, effective quarter, in-window indicator, and primary source.

Bundled cost-containment panel. We construct a new state-year panel (459 rows) coding five bundled actions per state-year: MAC list updated, PDL refreshed, NADAC alignment, pharmacy carve-out active, single-PBM transition. Coding draws on NASHP, HMA, KFF Medicaid Budget Surveys, and state Medicaid bulletins; every coded cell retains a source URL. A composite `bundle_count` sums the five indicators. MAC-list updates are the weakest dimension because we code only overhauls tied to spread-pricing enforcement, likely under-coding routine updates; PDL-refresh coding relies on uniform-PDL adoption; NADAC-alignment relies on statutory or contractual NADAC-benchmark requirements. The composite DiD is robust to moderate misclassification in any single indicator because power comes from the aggregate.

Outcomes and analytic sample. The primary outcome is log gross MCO generic spending per MCO enrollee per quarter (we use “log points” throughout; a coefficient of -0.058 is approximately -5.6 percent, and -0.252 is approximately -22.2 percent, computed as $1 - \exp(\text{coef})$). Secondary outcomes: log gross MCO brand spending per MCO enrollee (triple-difference placebo), log FFS generic spending per FFS enrollee (negative control), MCO generic NADAC markup, and channel totals with no denominator. Treatment variables are `spread_ban_enacted` (binary in-force indicator) and `cohort_year/cohort_quarter` (CS-DiD cohort indices). The analytic panel has 1,836 state-quarter observations; the primary outcome has 1,411 non-missing cells after the SDUD–NADAC–enrollment merge. We use “spread-pricing ban” consistently throughout (not “spread-pricing prohibition”), matching NASHP terminology.

5. Empirical Strategy

Primary estimator. We estimate Callaway–Sant’Anna20 staggered DiD with a never-treated comparison group and `differences` (v0.2.0) defaults, aggregating group-time average treatment effects $\text{ATT}(g, t)$ to an overall ATT and an event-time $\text{ATT}(k)$. We report pointwise 95 percent CIs on the primary event study and a simultaneous sup- t band over the post-period computed via an equicorrelation approximation ($\rho = 0.3$, $n_{\text{boot}} = 2,000$).

TWFE benchmark and Goodman-Bacon decomposition. Alongside CS-DiD we report a TWFE benchmark:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot \text{Post}_{st} + \varepsilon_{st}$$

with state-clustered standard errors. Because TWFE in staggered settings can place non-trivial weight on late-versus-early (already-treated-as-control) comparisons, we report a Goodman-Bacon decomposition21 on the balanced panel. We report the TWFE point estimate alongside the Bacon-clean weighted average (excluding late-vs-early comparisons) and the weight share on contaminated cells. **Standard errors for the Bacon-clean weighted average** are from a 500-replication cluster (state) nonparametric bootstrap: we resample the 36 balanced-panel states with replacement, re-run the full Bacon decomposition on each replicate, recompute the clean-comparison weighted average, and take the standard deviation across replicates. CS-DiD is our primary estimator; TWFE and Bacon-clean are benchmarks.

Why CS-DiD and TWFE standard errors differ. The CS-DiD standard error (SE 0.477 on the simple aggregate) is approximately double the TWFE SE (0.231) because the CS-DiD SE integrates (i) doubly-robust-propensity-score adjustment, (ii) a multiplier bootstrap over the $\text{ATT}(g, t)$ cells, and (iii) uncertainty over the never-treated control group weights. The TWFE SE clusters only at the state level. Neither SE is “correct”; they answer different questions. CS-DiD SEs are appropriate for inference on a cohort-heterogeneous ATT; TWFE SEs are appropriate for inference on a fixed-coefficient weighted sum. We report both transparently. In the event study we use CS-DiD’s analytic SEs (which are smaller than the overall-aggregate SE because event-time cells average over fewer unrelated cohort-calendar combinations).

Triple-difference (generic versus brand within MCO). We stack the MCO panel by drug type and estimate

$$\ln(\text{spend}_{std}) = \alpha_{sd} + \gamma_{td} + \beta \cdot (\text{Post}_{st} \times \text{generic}_d) + \delta \cdot \text{Post}_{st} + \varepsilon_{std}$$

where α_{sd} is state-by-drug-type and γ_{td} is quarter-by-drug-type fixed effects. The triple coefficient absorbs any state-quarter shock that moves generics and brands equally; what remains is generic-specific. Under the reframe, the triple-difference is evidence of *generic-targeted* state action within

MCO — consistent with either spread-pricing bans operating on generics or with bundled cost-containment levers (MAC updates, PDL refreshes, NADAC alignment) that hit generics more than brands. The design cannot separate these interpretations on its own.

Negative control (MCO versus FFS). We stack by channel:

$$\ln(\text{spend}_{stc}) = \alpha_{sc} + \gamma_{tc} + \beta \cdot (\text{Post}_{st} \times \text{MCO}_c) + \delta \cdot \text{Post}_{st} + \varepsilon_{stc}$$

Under the clean-mechanism hypothesis, β is negative (MCO responds, FFS does not). We pre-registered the FFS measurement drill (script `06_ffs_measurement_drill.py`) to determine whether any FFS effect is mechanical or substantive.

Bundled-action DiD. On the state-year panel with state and year fixed effects and state-clustered SEs:

$$y_{st}^{(k)} = \alpha_s + \gamma_t + \beta^{(k)} \cdot \text{SpreadBanInForce}_{st} + \varepsilon_{st}$$

for k indexing MAC updated, PDL refreshed, NADAC aligned, pharmacy carve-out active, single-PBM transition, and bundle_count. A positive $\beta^{(k)}$ says ban-enacting states disproportionately pursued the bundled action.

Case-study event studies. Ohio SPBM (October 2022): Ohio plus eight ban-only peers (AR, FL, MD, MA, NY, NC, PA, VA; Kentucky excluded because of its July 2021 single-PBM). Medi-Cal Rx (January 2022): California plus 23 never-treated peers (excluding the 24 ban states and five structurally-always-carved states WV, MO, TN, WI, ND).

Structural bounds using the Ohio prior. We use the 2018 Ohio audit’s 31.4 percent generic spread as a calibration prior: under full pass-through, eliminating a 31.4 percent spread should reduce MCO generic spending by $\ln(1 - 0.314) = -0.377$ log points. For each candidate estimator, we invert the log-point coefficient to an implied cross-state-average spread fraction and its 95 percent CI. This lets us state a power result explicitly: the triple-difference CI is consistent with spread fractions between roughly -12.5 percent (i.e., an implausible small positive wedge for PBMs) and $+29.6$ percent.

HonestDiD bounds. We report Rambachan–Roth23 relative-magnitudes bounds on the triple-difference event study for $M \in \{0.5, 1.0, 2.0\}$.

Robustness. (i) Policy-window subsample dropping the 10-state 2018Q1 contract mass cohort; (ii) Vermont-included sensitivity (unchanged at -0.063 , SE 0.226); (iii) Tennessee-included sensitivity (unchanged at -0.054 , SE 0.239); (iv) drop-Michigan sensitivity (unchanged at -0.061 , SE 0.229, rules out the Michigan-negative-residual concern); (v) CS-DiD with alternative SE construction (analytic vs. multiplier bootstrap; see Appendix Table A4).

6. Results

6.1 Raw cohort trends and cohort support

Figure 1 plots raw log MCO generic spending per MCO enrollee by cohort as small multiples, with state-level within-cohort lines, cohort means, never-treated mean, and onset markers. The 2018Q1 mass cohort ($n=10$) dominates early identification; four cohorts (NY, AR, VA, MD, KY/PA, NC, MA, FL) are one- or two-state cohorts with correspondingly weak late-lag support. Figure 2 plots cohort size by effective quarter (Panel A) and the $\text{ATT}(g,t)$ support heatmap (Panel B), making explicit that event-time $k + 12$ is supported almost entirely by the 2018Q1 cohort. Table 1 reports pre/post means for every outcome by treatment status.

6.2 Main DiD estimates

Table 2 reports the main DiD. The CS-DiD simple aggregate is -0.054 log points (SE 0.477) on log MCO generic spending per MCO enrollee; TWFE is -0.058 (SE 0.231, $n = 1,409$). The two estimators agree on the point estimate but disagree on precision, for the reasons discussed in Section 5. Figure 3 shows the CS-DiD event study (pointwise and simultaneous sup- t bands over the post-period); every post-period $ATT(k)$ crosses zero. Appendix Figure A1 overlays CS-DiD with the TWFE event study benchmark. The MCO-NADAC markup outcome moves the wrong direction for a spread-pricing-reduction story ($+37.6$ percentage points, SE 31.2), though also not significant — consistent with NADAC markup being a contemporaneous function of both state reimbursement policy and PBM pricing behavior. When the 10-state 2018Q1 contract cohort is dropped, TWFE is -0.039 (SE 0.254). The full-panel TWFE estimate drawing only on never-treated controls is $+0.067$ (SE 0.233) — wrong-signed, indicating that the never-treated comparison group differs from treated in ways that the full panel’s not-yet-treated comparisons help correct; this is the empirical motivation for CS-DiD’s preference for not-yet-treated comparisons where available.

6.3 Goodman-Bacon decomposition

Table 3 and Figure 4 report the Goodman-Bacon decomposition of the TWFE benchmark on the balanced 36-state panel. Total weight on contaminated late-versus-early comparisons is 12.6 percent, with a weighted-average estimate of $+0.323$ — positive-signed, consistent with contamination from late-cohort treatment effects evaluated against already-treated early cohorts whose post-period is itself trending. The clean weighted average (Never-vs-timing and Early-vs-Late together) is **-0.118 log points (cluster-state bootstrap SE 0.295, 500 replications; 95 percent percentile CI $[-0.735, +0.470]$)**, nearly identical to the triple-difference point estimate of -0.117 . The Bacon-clean estimate is the TWFE benchmark with the contamination stripped out; its point estimate and sign strengthen the triple-difference’s direction, but the bootstrap-derived CI is substantially wider than the TWFE benchmark SE (0.231) because it captures the state-resampling variance in the weighted combination of 2×2 cells rather than treating the weights as fixed. The bootstrap SE is the methodologically correct object for inference on the Bacon-clean quantity, and we use it in place of the TWFE-SE proxy used in earlier drafts.

6.4 Triple-difference

Table 4 reports the triple-difference. In the full sample, $\text{Post} \times \text{generic} = -0.117$ log points (SE 0.120, $t = -0.97$, $n = 2,793$); in the policy-window subsample, -0.149 (SE 0.147). The brand placebo passes cleanly ($+0.059$, SE 0.290 full sample). The triple-difference SE is approximately half the generic-only DiD SE, reflecting state-quarter absorption. **Interpretation of the triple-difference under the reframe.** The triple-difference absorbs state-quarter common shocks but not state-quarter generic-specific shocks. Under the bundled cost-containment reading, bundled actions (MAC, PDL, NADAC alignment) hit generics more than brands because generics are where state cost-containment levers operate. The triple-difference is therefore *evidence that state action during the ban window was generic-targeted*, not independent evidence for the spread-pricing mechanism. A PBM-skeptical reading and a cost-containment-bundle reading are both consistent with a negative triple-difference and a passing brand placebo. Section 6.7 uses the Ohio prior to quantify the fraction of the triple-difference coefficient consistent with each interpretation.

6.5 Negative control and FFS measurement drill

Table 5 reports the negative control. **The negative control fails.** The DiD coefficient on log FFS generic spending per FFS enrollee is -0.252 (SE 0.205, $n = 1,752$) in the full sample and -0.289 (SE 0.258) in the policy-window subsample. Both are larger in magnitude than the MCO coefficient. The MCO-versus-FFS triple-difference is $+0.195$ (SE 0.361) — wrong-signed for a spread-pricing mechanism.

Table 5b reports the pre-registered FFS measurement drill. None of the following specifications eliminate the FFS effect: dropping states with mean FFS enrollment below 25,000 / 50,000 / 100,000 / 250,000; dropping states with any quarterly FFS enrollment below 1,000; winsorizing per-enrollee values at the 95th or 99th percentile; using a common total-Medicaid-enrollment denominator; and running a channel-total specification with no denominator at all. In every specification, the MCO estimate is a precise null (between $+0.04$ and -0.06 log points) while the FFS estimate remains between -0.17 and -0.33 . **The channel-total specification is decisive.** At the scale of total FFS generic spending (no denominator), the decline is -0.308 (SE 0.199) for FFS and $+0.043$ (SE 0.213) for MCO. The “channel-total” label denotes the no-denominator specification; because the common-denominator specification delivers the same asymmetric pattern, enrollment-driven noise cannot drive the channel-total result.

The FFS effect is substantive. State-level generic spending declines in treated states are real and present in both channels. The clean-mechanism story would require the FFS channel to be zero. It is not. What the channel-total and measurement drill do *not* rule out is that NDC-level composition in the FFS channel was shifting in treated states during the post-period for reasons unrelated to the ban (e.g., a few high-cost specialty generics being dropped from the FFS channel), or that treated states were a biased sample in terms of pre-period FFS trends (the pre-period variation the HonestDiD bounds acknowledge).

6.6 Bundled cost-containment evidence

Table 6 and Figure 5 report the bundled-action DiDs on the 459-row state-year panel. The composite bundle-count coefficient on the spread-pricing-ban-in-force indicator is **$+0.507$ (SE 0.107, $p < 0.001$)**: treated states take approximately half an additional bundled cost-containment action per state-year beyond state and year fixed effects. Three of five action-specific coefficients are significant: MAC list updated **$+0.199$ (SE 0.039, $p < 0.001$)**, NADAC alignment **$+0.176$ (SE 0.041, $p < 0.001$)**, PDL refreshed **$+0.089$ (SE 0.039, $p = 0.026$)**. Single-PBM transition is marginal ($+0.031$, SE 0.018, $p = 0.087$). Pharmacy carve-out active is a precise zero ($+0.012$, SE 0.031, $p = 0.705$), consistent with carve-outs being a substitute for spread-pricing bans (West Virginia, Missouri, Tennessee, Wisconsin, North Dakota carved out rather than ban).

The bundled-action evidence is the empirical foundation of the reframe. MAC-list updates, NADAC alignment, and PDL refreshes affect pharmacy prices in *both* MCO and FFS channels through shared state pharmacy administrative infrastructure. The joint MCO-plus-FFS decline in treated states is the predicted signature of a bundle rather than a failure of a clean negative control. Under this reading, the triple-difference’s negative (but not significant) point estimate reflects generic-targeted state action during the ban window — which could be spread-pricing, bundled cost containment, or both.

6.7 Structural bounds using the Ohio prior

Table 7 reports the structural decomposition. Full pass-through of the Ohio 31.4 percent generic spread would imply $\ln(1 - 0.314) = -0.377$ log points on MCO generic spend per enrollee. The triple-difference point estimate -0.117 implies a cross-state-average generic spread of **11.0 percent**, with a 95 percent CI of **[−12.5 percent, +29.6 percent]**. The Ohio 31.4 percent prior falls outside the 95 percent CI. The upper edge of the CI is consistent with a cross-state-average spread fraction near 30 percent — close to Ohio but not as large. The TWFE CI implies a spread band of **[−48.5 percent, +40.0 percent]** (wider because TWFE is noisier). The Bacon-clean weighted-average estimate implies 11.1 percent mid-point spread with a cluster-bootstrap 95 percent spread CI of **[−58.5 percent, +50.1 percent]** (SE 0.295 on the log-point coefficient) — the widest of the three because the bootstrap captures state-cluster resampling variance in the 2×2 weighted combination; it cannot reject the Ohio prior. Figure 6 plots the three estimates against the Ohio prior on a common axis. The decisive test — whether the cross-state-average spread is above or below 15 percent — requires either claim-level T-MSIS data or the spread-reporting infrastructure from the November 2023 Medicaid Managed Care final rule, which will come online in rating periods beginning November 2025.

6.8 Case studies: Ohio SPBM and Medi-Cal Rx

Ohio SPBM (October 2022). Ohio moved to a single state-contracted PBM on October 1, 2022. Ohio’s Milliman actuarial evaluation reported \$140 million in savings over two years using internal accounting that separately attributes dispensing fees, PBM administrative fees, and retained spread. In SDUD, the Ohio SPBM coefficient on log MCO generic spending per MCO enrollee is $+0.158$ (SE 0.369, $p=0.680$) — a precise zero; the ban-only comparison pool is itself bundled, and with a single treated unit and at most eight post-SPBM quarters, cluster-robust inference is weak.

Medi-Cal Rx (January 2022). California moved MCO pharmacy volume into Magellan on January 1, 2022. The channel flips are dramatic: FFS per-enrollee $+1.882$ log points (SE 0.313, $p<0.001$) as volume moves to FFS while FFS enrollment does not grow proportionally; MCO per-enrollee -2.757 (SE 0.099, $p<0.001$); **total generic per total Medicaid enrollee $+0.189$ (SE 0.059, $p=0.004$)** — a modest precisely estimated *increase*. Contemporaneous generic inflation, continuous-coverage unwinding, 340B interactions, and documented 2022 implementation disruptions all contribute. A structural, well-identified pharmacy carve-out produced no visible aggregate savings in SDUD in its first two post-period years.

Both case studies deliver the data-infrastructure finding: public SDUD aggregates cannot cleanly evaluate structural pharmacy benefit reform on short post-period horizons because gross-amount-reimbursed does not isolate PBM-retained spread and because contemporaneous shocks to the denominator are only partially absorbed.

6.9 HonestDiD bounds

Table 8 reports the relative-magnitudes pre-trend sensitivity bounds on the triple-difference event study. The maximum absolute pre-period coefficient is 0.378 log points. At $M = 0.5$, the bias bound is 0.189, and every honest 95 percent CI already contains zero for post-period quarters 0 through 11; at $M = 1.0$ and $M = 2.0$ the bounds widen further. **Power statement.** The data cannot distinguish “bans reduced MCO generic spending by 10 to 30 percent as Ohio predicted” from “bans had zero effect”; even under a modest $M = 0.5$ relative-magnitudes restriction, zero is not excluded at any post-period quarter.

7. Discussion

7.1 What the data support and the committed reframe

The committed reframe: **the pre-registered clean identification strategy failed; the public-data panel cannot distinguish spread-pricing from contemporaneous state pharmacy cost-containment; the MCO point estimate is consistent with anywhere from substantial savings (lower edge of the triple-difference CI -30 percent) to zero; and the decisive test awaits claim-level data.** The reframe is not an apology for the failed negative control — it is a substantively grounded reading anchored by three empirical objects: the bundled-action panel ($+0.507$, $p < 0.001$), the Bacon-clean decomposition (contaminated share 12.6 percent; clean weighted average -0.118), and the structural bounds (11.0 percent implied spread, 95 percent CI $[-12.5, +29.6]$ percent against a 31.4 percent Ohio prior). The triple-difference is evidence of generic-targeted state action during the ban window; under the bundle reading, much or most of that action is non-spread-pricing cost-containment operating through shared pharmacy infrastructure. This does not mean spread-pricing bans have zero effect; it means public SDUD/NADAC aggregates cannot separately identify the spread-pricing contribution from the bundle.

7.2 Implications for state and federal policy

For state Medicaid directors, the strongest actionable finding is that bans are empirically embedded in a bundle. A state acting on spread pricing alone — without MAC updates, PDL refreshes, or NADAC alignment — is likely to see smaller aggregate savings than a state undertaking the full bundle. The directional evidence supports bundled adoption; the precise spread-pricing contribution cannot be cleanly identified from public data.

For federal policymakers implementing the CAA 2026 Medicare Part D spread-pricing prohibition, the paper delivers a data-infrastructure warning. The spread-reporting requirement in the November 2023 Medicaid Managed Care final rule is the right first step; it will produce state-level aggregate data on the MCO-pharmacy wedge beginning in the 2025–2026 rating period. Federal evaluators should establish an analogous claim-level data infrastructure for Part D before the 2028 effective date, so that a proper negative control is available when the ban takes effect. Without such infrastructure, the Part D evaluation will face the same identification failure this paper documents for Medicaid.

7.3 Why publish the public-data paper now rather than wait for T-MSIS

The paper establishes, independently of any forthcoming claim-level analysis, that public-data methods cannot resolve the mechanism question. That is a methodological contribution to the applied Medicaid policy literature and a direct implication for federal evaluation design. A companion paper using claim-level T-MSIS data (access application pending) will resolve the mechanism directly. Publishing now establishes the negative-control failure, the bundled-action panel, and the data-infrastructure implications at the moment when those implications are most policy-salient (post-CAA 2026, pre-Part D effective date).

7.4 Limitations

Aggregation level. SDUD reports state-quarter-NDC aggregates; the MCO-payment-minus-pharmacy-reimbursement spread cannot be directly observed from public data. **Contemporaneous policies.** The failed negative control and the bundled-action panel document that MAC,

PDL, and NADAC-alignment actions are material confounders; isolating the spread-pricing mechanism from the bundle requires data the public-data panel does not have. **Few treated clusters per cohort.** Several cohorts include one state; cluster-robust inference with few treated clusters per cohort may be mildly anti-conservative. **Parallel trends.** Deep-lead pre-period variation is visible; HonestDiD bounds at $M = 0.5$ already contain zero for every post-period quarter. **Case-study power.** Ohio SPBM and Medi-Cal Rx have one treated unit each and short post-period windows. **Bundled-panel coding.** MAC-list coding is the weakest dimension because we code only overhauls explicitly tied to spread-pricing enforcement. **T-MSIS access pending.** The decisive mechanism test is out of scope.

8. Conclusion

State Medicaid spread-pricing bans during the 2018–2024 wave did not produce statistically detectable changes in MCO generic drug spending per enrollee at the state-quarter aggregate level in public CMS data. CS-DiD is -0.054 log points (SE 0.477); TWFE is -0.058 (SE 0.231); the Bacon-clean weighted average is -0.118 ; the triple-difference is -0.117 (SE 0.120) with a clean brand placebo; and the pre-registered MCO-versus-FFS negative control fails — FFS generic spending declines by -0.252 log points (SE 0.205) in treated states, substantively rather than mechanically. A new state-year panel of five bundled pharmacy cost-containment actions shows that treated states take approximately half an additional bundled action per state-year beyond state and year fixed effects ($+0.507$, SE 0.107, $p < 0.001$), concentrated in MAC updates, NADAC alignment, and PDL refreshes. A structural decomposition using the 2018 Ohio 31.4 percent audit prior places the implied cross-state-average generic spread from the triple-difference at 11.0 percent, 95 percent CI $[-12.5, +29.6]$ percent — statistically below Ohio but consistent with anywhere from zero to nearly-Ohio-sized.

The honest bottom line. The null here is not evidence that spread-pricing bans fail. It is evidence that public data cannot be used to cleanly test the spread-pricing mechanism in isolation from contemporaneous bundled cost containment. The decisive mechanism test awaits claim-level Medicaid data or the spread-reporting infrastructure that will begin under the November 2023 Medicaid Managed Care final rule. For the imminent federal Medicare Part D prohibition under CAA 2026, the paper’s clearest contribution is a warning: federal evaluators must establish a claim-level data infrastructure now, so that when the 2028 effective date arrives, a proper negative control is available.

Tables and Figures

Table 1. Summary statistics on the state-quarter panel, pre- and post-period means by treatment status. Source: `analysis/tables/summary_statistics.csv`.

Table 2. Main DiD estimates on log MCO generic spending per MCO enrollee. Columns: TWFE, Callaway–Sant’Anna. Rows: full sample, policy-window, Vermont-included, Tennessee-included, drop-Michigan, never-treated-only. Source: `analysis/tables/main_did_summary.csv`, `analysis/tables/policy_window_summary.csv`.

Table 3. Goodman-Bacon decomposition of the static TWFE benchmark. Columns: 2×2 comparison type, weight, average estimate. Rows: Never-vs-timing, Early-vs-Late, Late-vs-Early. Source: `analysis/tables/goodman_bacon_decomp.csv`.

Table 4. Triple-difference (generic versus brand within MCO). Columns: generic-only, brand-only placebo, triple-difference. Rows: full sample and policy-window subsample. Source: `analysis/tables/triple_diff_summary.csv`.

Table 5. Negative control (MCO versus FFS). Columns: FFS DiD, MCO-versus-FFS triple-difference. Rows: full sample, policy-window. Source: `analysis/tables/negative_control_summary.csv`.

Table 5b. FFS measurement drill. Rows: drop mean FFS below 25k/50k/100k/250k; drop any quarter FFS below 1,000; winsorize at p95/p99; common total-Medicaid denominator; channel-total (no denominator); log-vs-level comparison. Source: `analysis/tables/ffs_measurement_drill.csv`.

Table 6. Bundled cost-containment DiD. Outcomes: five action-specific indicators and composite bundle-count. TWFE with state and year fixed effects, state-clustered SEs. Source: `analysis/tables/bundled_cost_containment_summary.csv`.

Table 7. Structural bounds using the 2018 Ohio audit prior. Rows: TWFE, triple-difference, Bacon-clean weighted average. Columns: log-point coefficient, 95 percent CI, implied cross-state-average spread fraction, Ohio prior rejection indicator. Source: `analysis/tables/structural_bounds.csv`.

Table 8. HonestDiD relative-magnitudes bounds on the triple-difference event study for $M \in \{0.5, 1.0, 2.0\}$. Source: `analysis/tables/honestdid_bounds_summary.csv`.

Figure 1. Raw log MCO generic spend per MCO enrollee by cohort (small multiples). Source: `analysis/figures/cohort_small_multiples.png`.

Figure 2. Cohort support: (A) state count by effective quarter, (B) ATT(g,t) support heatmap. Source: `analysis/figures/cohort_support.png`.

Figure 3. CS-DiD event study for log MCO generic spend per enrollee with 95 percent pointwise and sup- t simultaneous bands. Source: `analysis/figures/csdid_event_study_ln_gen_spend_mco_uniform.png`.

Figure 4. Goodman-Bacon decomposition scatter (weight $\times$ 2 $\times$ 2 estimate). Source: `analysis/figures/goodman_bacon_scatter.png`.

Figure 5. Bundled cost-containment DiD coefficients with 95 percent CIs. Source: `analysis/figures/bundled_cost_containment_coefs.png`.

Figure 6. Structural bounds: observed estimates vs. Ohio-audit prior. Source: `analysis/figures/structural_b`.

Appendix Figure A1. CS-DiD vs. TWFE event-study overlay. Source: `analysis/figures/csdid_vs_twfe_ove`.

Appendix Figure A2. FFS measurement drill: MCO and FFS DiD coefficients across exclusion thresholds. Source: `analysis/figures/ffs_drill_by_threshold.png`.

Appendix Figure A3. HonestDiD bounds (full sample and policy-window). Source: `analysis/figures/honestdid_bounds_full.png`, `analysis/figures/honestdid_bounds_drop_2018q1.png`.

Appendix Figure A4. Ohio SPBM event study. Source: `analysis/figures/ohio_spbm_event_study.png`.

Appendix Figure A5. Medi-Cal Rx event study. Source: `analysis/figures/medi_cal_rx_event_study.png`.

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